NAME : SHWETA

SRN : PES1UG24AM404

PROJECT N0: 40

SE(LAB1)

PROBLEM STATEMENT #40 |

Retail, E-Commerce & Finance Student Startup Crowdfunding Platform

1. Problem Context & Overview

A crowdfunding engine for university student ventures, supporting milestone-based funding targets, backer reward tier selection, and verified fund releases upon faculty committee review.

Target Stakeholders / Actors: Student Founder, Campaign Backer

2. Functional Requirement (FR)

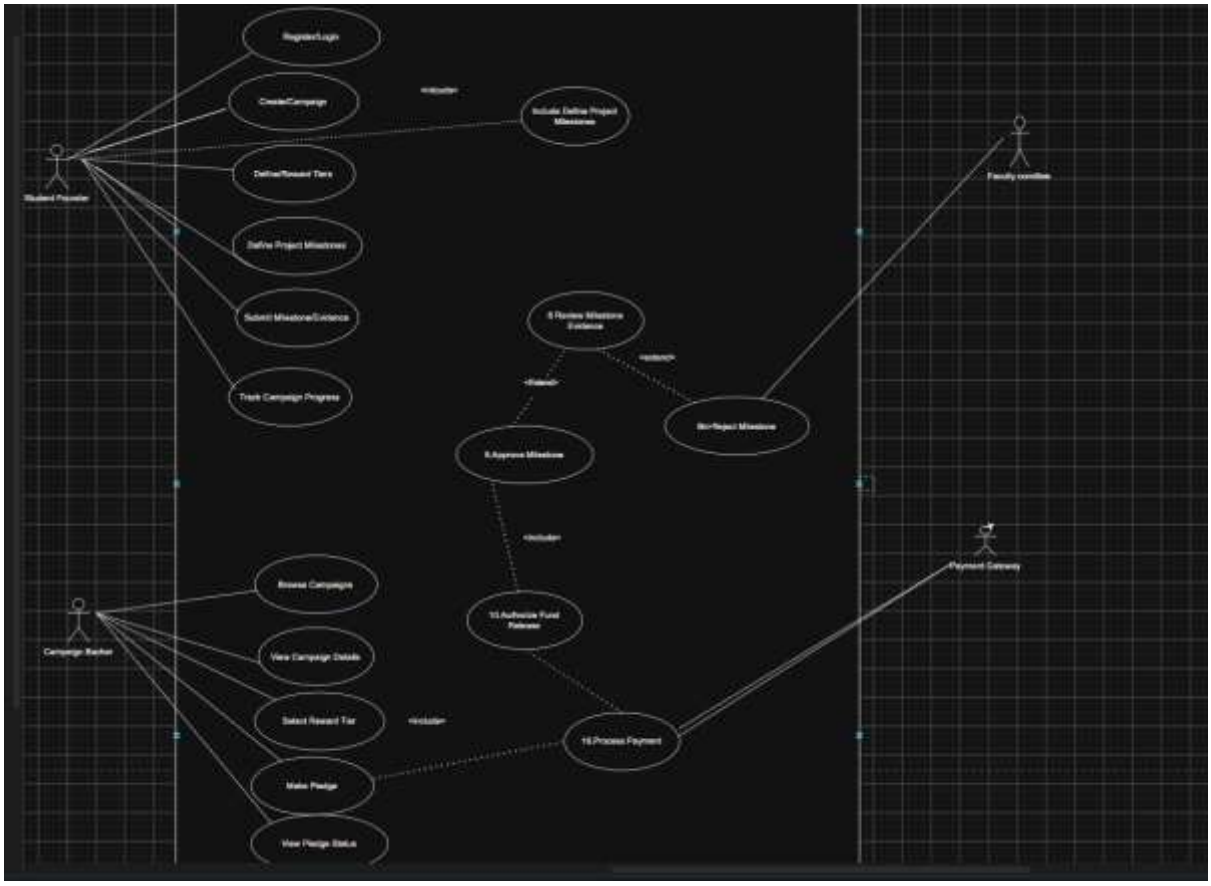
ID	Type	Description	Priority	Rationale	Acceptance Criteria
FR-001	Functional	The system shall allow a **Student Founder** to create a crowdfunding campaign by providing the startup details, funding target, campaign duration, reward tiers, and project milestones.	High	Allows student founders to present their ventures and define their funding requirements.	**Pass:** Campaign is created successfully with all mandatory details. **Fail:** Campaign is created with missing or invalid mandatory information.
FR-002	Functional	The system shall allow **Campaign Backers** to browse available student campaigns and select a suitable reward tier.	High	Allows student founders to present their ventures and define their funding requirements.	**Pass:** Backer can view campaign details and successfully select an available reward tier. **Fail:** Unavailable or invalid reward tier cannot be selected.

FR-003	Functional	The system shall allow **Campaign Backers** to make financial pledges to active crowdfunding campaigns through the payment gateway.	High	Provides the core crowdfunding functionality for collecting financial support.	**Pass:** Valid pledge is processed and the campaign's collected amount is updated. **Fail:** Invalid payment or pledge is not recorded.
FR-004	Functional	The system shall allow **Student Founders** to submit evidence for completed project milestones for review by the Faculty Committee.	High	Provides evidence that project milestones have been completed before funds are released.	**Pass:** Milestone evidence is successfully submitted and made available for faculty review. **Fail:** Incomplete or invalid evidence is not submitted
FR-005	Functional	The system shall allow **Student Founders** to submit evidence for completed project milestones for review by the Faculty Committee.	High	Ensures that campaign funds are released progressively and only after milestone verification.	**Pass:** Approved milestone results in the corresponding tranche being released. **Fail:** Funds are not released without faculty milestone approval.

3.Non Functional requirements:

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The payment processing gateway shall use secure tokenized payment authorization and comply with applicable **PCI-DSS** requirements.	High	**Pass:** Payment processing demonstrates secure tokenization and satisfies the defined security checks. **Fail:** Payment information is processed without the required security controls.	Protects backers' payment information and reduces the risk of unauthorized access or payment-data exposure.
NFR-002	Performance	The system shall provide responsive campaign browsing, pledge processing, and milestone-review operations under simulated peak user load	Medium	**Pass:** Performance testing under simulated peak load meets the defined response-time target. **Fail:** Response time exceeds the defined target or the system becomes unavailable.	Ensures the platform remains usable when many backers and student founders access it simultaneously.

UML DIAGRAM:



USE CASE FLOW:

USE CASE FLOW SPECIFICATION

Use Case: Verify Milestone and Release Funds
Primary Actor: Faculty Committee
Supporting Actors:
- Student Founder
- Payment Gateway

1. PRECONDITIONS
- A crowdfunding campaign exists.
- The campaign has predefined project milestones.
- The Student Founder has submitted milestone evidence.
- The Faculty Committee is authorized to review the milestone.
- The campaign has received sufficient pledges for fund release.

2. POSTCONDITIONS

- The milestone is marked as approved.
- The corresponding funding tranche is released.
- The campaign funding status is updated.
- The transaction is recorded.
- The Student Founder is notified.

3. MAIN SUCCESS SCENARIO

1. Student Founder submits milestone evidence.
2. System records the submitted evidence.
3. Faculty Committee opens the milestone for review.
4. System displays the milestone details and evidence.
5. Faculty Committee verifies the evidence.
6. Faculty Committee approves the milestone.
7. System authorizes the corresponding funding tranche.
8. Payment Gateway processes the fund transfer.
9. System updates the milestone and campaign status.
10. System sends a confirmation to the Student Founder.

4. ALTERNATE FLOW

A1. Milestone Evidence Rejected

1. Faculty Committee reviews the submitted evidence.
2. The evidence does not satisfy the milestone criteria.
3. Faculty Committee rejects the milestone and provides comments.
4. System marks the milestone as "Rejected".
5. System does not release the funding tranche.
6. Student Founder is notified and can submit additional evidence or take corrective action.